ModelIt!

Teacher's Guide

Your Inner Fish

Why You Share Bones with a 375-Million-Year-Old Fish

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-LS4-1, MS-LS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS4-1, MS-LS4-2

Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth. Apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern organisms and between modern and fossil organisms to infer evolutionary relationships.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze patterns in fossil evidence and anatomical data to construct explanations of evolutionary relationships.
Disciplinary Core Idea	**LS4.A Evidence of Common Ancestry and Diversity** — The fossil record documents the existence, diversity, extinction, and change of many life forms throughout history.
Crosscutting Concept	**Patterns** — Students identify patterns of anatomical similarity across species and through geological time as evidence of common ancestry.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist discovers that the flipper of a whale, the wing of a bat, and the arm of a human all contain the same bones. What does this evidence most likely indicate?

CONSTRUCTED RESPONSE:

Using evidence from the fossil record, explain how scientists know that modern land vertebrates evolved from ancient fish. Include at least two types of evidence.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The fossil record provides overwhelming evidence that life on Earth has changed dramatically over billions of years. By studying fossils and comparing anatomy, scientists reconstruct evolutionary relationships and understand how modern organisms are connected to ancient life.

The Fossil Record

Fossils form when organisms are buried in sediment before decomposition. Over millions of years, minerals replace organic material. The fossil record is incomplete — fossilization is rare — but it provides a clear timeline showing how life has changed across Earth's history.

Homologous Structures

The arm of a human, the wing of a bat, the flipper of a whale, and the leg of a dog all contain the same basic bones: humerus, radius, and ulna. These homologous structures have the same evolutionary origin but have been modified for different functions, providing strong evidence for common ancestry.

Tiktaalik: Your Inner Fish

In 2004, paleontologist Neil Shubin discovered Tiktaalik roseae, a 375-million-year-old fossil bridging fish and land animals. It had fish features (scales, fins) AND land features (flat head, neck, primitive wrist bones). It's one of the most famous transitional fossils ever found.

Extinction and Diversification

Earth has experienced at least five mass extinction events. The most famous killed the non-avian dinosaurs 66 million years ago. But extinctions also open ecological niches — after the dinosaurs disappeared, mammals diversified rapidly, eventually leading to primates and humans.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify environmental change rate, geological time, anatomical similarity, and species diversity as components of the evolutionary system.

E — Establish Relationships
Students determine that environmental change drives both extinction (reducing diversity) and adaptation (changing anatomy over time).

V — Visualize & Model
Students build a model showing how environmental pressures and time interact to produce changes in species diversity and body structure.

E — Evaluate Outcomes
Students run stable, mass extinction, and gradual change scenarios to observe different evolutionary outcomes.

R — Revise & Extend
Students add mutation rate or geographic isolation to explore more realistic evolutionary dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at your hand. Now look at this fish from 375 million years ago. Same bones."*

Watch For

Show side-by-side image of Tiktaalik and human arm bones.

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating evidence that connects every animal alive to ancient ancestors."*

Watch For

Have students read objectives. Pre-teach 'homologous structure.'

Slide 3: Big Question

3 min

Say: *"This isn't coincidence. This is evidence of something incredible."*

Watch For

Show images of homologous structures. Ask: Why the same bones?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how the environment shapes life over millions of years."*

Watch For

Preview LEVER steps. Connect to prior knowledge about fossils.

Slide 5: Activity 1: Components

8 min

Say: *"What factors drive evolutionary change? What can the environment control?"*

Watch For

Guide sorting. Discuss why time and environment are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When the environment changes rapidly, what happens to species diversity?"*

Watch For

Help students identify relationships. Discuss dual effects.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's simulate a mass extinction and see what our model predicts!"*

Watch For

Students predict first. Compare stable vs. mass extinction scenarios.

Slide 8: Discoveries

5 min

Say: *"How does the fossil record match our model? What does Tiktaalik prove?"*

Watch For

Show Tiktaalik fossil. Connect to model predictions.

Slide 9: STEM Challenge

12 min

Say: *"You're now museum designers. Build an exhibit showing evidence for evolution."*

Watch For

Groups design evidence walls using fossil data.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Environmental Change Rate, Time (Millions of Years)
- **INTERNAL:** Anatomical Similarity, Species Diversity

Relationships

- **Environmental Change Rate to Species Diversity:** NEGATIVE (-) — Rapid environmental change causes extinctions, reducing the number of species. Organisms that can't adapt fast enough go extinct.
- **Time (Millions of Years) to Anatomical Similarity:** NEGATIVE (-) — Over longer time periods, species diverge more from common ancestors, reducing anatomical similarity between distant relatives.
- **Species Diversity to Anatomical Similarity:** POSITIVE (+) — Higher species diversity means more organisms share recent common ancestors, retaining more similar body structures.

Key Discoveries

- Species that can't adapt to environmental change go extinct — creating gaps in the fossil record
- Homologous structures prove that very different-looking animals evolved from common ancestors
- Mass extinction events reduce diversity temporarily but open ecological niches for new species
- The fossil record shows a clear pattern: simple organisms came first, complexity increased over time

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Humans evolved from monkeys"	Humans and modern apes share a common ancestor from about 6-7 million years ago. We didn't evolve FROM chimpanzees — we're evolutionary cousins. Both species have been evolving separately.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Evolution is just a theory"	In science, 'theory' means a well-tested explanation supported by massive evidence. Evolution is supported by fossils, genetics, anatomy, and direct observation. Gravity is also 'just a theory.'
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fossils should show every step"	Fossilization is extremely rare — most organisms decompose completely. The fossil record will always have gaps, but the fossils we DO have consistently support evolutionary patterns.